

GNCAirstrike PN Upgrade Summary

GNCAirstrike — PN Guidance Upgrade Summary

Folder:

C:\Users\Meet\OneDrive\Desktop\Sem_9\Claude_Missile_guidance\GNCAirstrike-master **Files changed:** model.m, simulink_model.slx

What changed

1. Proportional Navigation guidance (replaces static FPA command)

The GUIDANCE COMMAND MATLAB Function chart was rewritten (function renamed FPA → PN GUIDANCE). It now computes: - LOS angle $\lambda = \text{atan2}(dh, d)$, recomputed every guidance tick from current position/target (haversine geometry, unchanged math). - LOS rate $\dot{\lambda}$ and closing velocity $V_c = -d\text{Range}/dt$, both via persistent-state finite differences. - $a_{\text{cmd}} = N_{\text{prime}} * V_c * \dot{\lambda}$ (true PN law). - Converted to the outer loop's actual reference domain by integrating a_{cmd} / V_m over time into a commanded flight-path angle (persistent γ_{cmd}), since the existing outer P-loop compares an **angle** (FPA), not a raw acceleration — this keeps the inner LQR/Kalman loop and its gains completely untouched. - The chart's sample time was set to a fixed discrete 0.1 s (was inherited from the variable-step ode45 solver) so the finite-difference state advances on clean, monotonic ticks. - N_{prime} (navigation constant, default **3.5**) is now a tunable variable in model.m.

2. Terrain/obstacle override — preserved, now on top of PN

The inline override in the chart ($D_{\text{OBS}} < 2000 \text{ m} \rightarrow$ force the guidance output) is kept exactly as a safety layer, now forcing a **5° climb** (this task's spec) instead of the as-found 0 rad, and the PN integrator state is held (not advanced) while the override is active so guidance resumes smoothly once past the obstacle.

3. Two pre-existing bugs found and fixed (both outside the guidance)

law itself, discovered because the closed loop would not converge to the target at all under either the original or the new guidance): - **Kinematics sign bug:** the dz/dt (altitude) integrator was missing a sign flip needed for the NED "Down" convention used elsewhere in the model ($P05 > LLA$ is an Aerospace Blockset FlatEarth2LLA block). Fixed with a single new Gain block (-1) on that one line — dx/dt untouched. - **Outer-loop gain too aggressive:** the outer P-only loop's output is added directly as a bias to the elevator command ($\delta_c = q_{\text{cmd}} - K \cdot \hat{x}$); given the plant's $B(2) = -331.4$, the original $P = -1.0$ overdrove the fin and diverged. Retuned **magnitude only** (sign unchanged) to $P = -0.1775$, derived analytically from the closed-loop DC gain and a 5 s target time constant. **LQR gains and Kalman filter design were not touched** — regression-verified: $\text{eig}(A-BK) = [-2.5132, -150.1351]$, observer eigenvalues $[-18.2152, -15.9376]$, identical before and after.

Verification (single trial, PN guidance)

- Simulation self-terminates via the model's own Stop block at **t ≈ 77.9 s** (matches the ~77 s nominal-flight-time estimate).
- Final altitude **784 m** (target elevation 795 m); final slant range ≈ **542 m**.
- Terrain override fired correctly during the same run (min distance to obstacle ≈ 33 m), confirming the 2 km/5° safety layer works on top of PN.

Not completed in this pass (time-boxed)

The Monte Carlo harness (monte_carlo.m, N=300 trials via Simulink.SimulationInput + parsim) and RESULTS.md were designed (variables to randomize identified: LAT_INIT/LON_INIT, LAT_TARGET/LON_TARGET with yaw_init recomputed consistently per trial; process/measurement noise via the w/noise/v/noise Band-Limited White Noise blocks' seed parameter) but not yet built/run in this session — flagged as the next step rather than rushed.